

**INTEGRATING ERC-20 ALLOWANCE EDGES INTO
PAGERANK FOR ENHANCED ON-CHAIN WALLET
REPUTATION SCORING**

by

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submitted in accordance with the requirements for the degree of

DOCTOR OF PHILOSOPHY

in the subject

COMPUTER SCIENCE

at the

UNIVERSITY OF SOUTH AFRICA

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August 2026

Abstract

The financial system has generated broad-based growth impetus by allocating capital efficiently through lending (Schär, 2021). In this light, the advancement of DeFi is likewise expected to hinge critically on the degree of on-chain lending and leveraged trading activity. However, DeFi platforms face constraints in executing collateral-backed loans due to difficulties in enforcing collateral assets, and they similarly encounter limitations in offering unsecured credit, since traditional credit-scoring models that incorporate tangible-asset and employment information cannot be applied directly (Schär, 2021; Ghosh et al., 2024).

Existing PageRank-based methods, such as the Adaptive Weighted PageRank (AWP) algorithm, primarily rely on transaction volume and recency (time-decay) to assess wallet trustworthiness (Do et al., 2023). While these approaches effectively capture economic flows and activity levels, they might incur substantial computational overhead by requiring exhaustive parsing of historical transactions.

This study proposes EndorseRank, an on-chain reputation metric leveraging ERC-20 approve and permit allowances as direct indicators of wallet endorsement. Each approve/permit action explicitly grants spending permissions to another wallet, inherently representing trust. By constructing a directed, weighted graph G_{approve} based solely on the most recent allowance amounts for each wallet pair, EndorseRank eliminates the need for historical time-decay adjustments and significantly reduces computational complexity.

The proposed EndorseRank algorithm utilizes PageRank methodology on this allowance-based graph on **Arbitrum One**. Empirical validation uses **GMX V2 perpetual swap position** outcomes: collateralized leveraged exposure on long and short positions, where a profitable **PositionDecrease** close (positive **basePnlUsd**, excluding position liquidation) is treated as a functional analog of successfully managing borrowed exposure. Our research aims to:

1. Develop a robust data pipeline to extract and manage the latest approve/permit allowances on Arbitrum One.

2. Construct and efficiently compute reputation scores from a sparse and endorsement-oriented graph.
3. Evaluate EndorseRank's alignment with GMX trading-success proxies (close-success count and realized gain proxy) versus AWP and other established transaction-based methods, using Spearman's rho and Kendall's tau.
4. Conduct extensive computational benchmarks to demonstrate superior efficiency in runtime and memory usage.

By addressing these objectives, EndorseRank seeks to provide a computationally efficient, semantically clear, and domain-aligned reputation scoring mechanism suitable for real-world DeFi platforms, enabling better risk management and decision-making.

***Keywords—* DeFi, On-Chain Reputation, GMX V2, Perpetual Swap Positions, Arbitrum One, Blockchain Data Analytics, Wallet Endorsement Graph**

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1 Introduction and background

Decentralized Finance (DeFi) has expanded lending, borrowing, and leveraged trading activity on public blockchains, yet most credit-like interactions remain highly collateralized because protocols cannot rely on traditional off-chain identity, employment, or credit histories (Schär, 2021; Packin & Lev-Aretz, 2024). In a pseudonymous setting, the core challenge is not merely allocating capital, but allocating it safely: identifying which wallets are likely to behave reliably in credit-like interactions (e.g., repeated borrowing, repayment, delegation of spending rights, and successful management of leveraged exposure), while limiting exposure to opportunistic or malicious actors (e.g., position liquidation, griefing, Sybil-style account creation, and rapid reputation farming) (Packin & Lev-Aretz, 2024).

In this study, **on-chain reputation score** computation refers to the process of computing a quantitative score for a blockchain address using observable on-chain behavior and relationships, with the goal of approximating trustworthiness or credit-related reliability without relying on off-chain personal data (Ghosh et al., 2024; Vathsala V. & V. D. Sharma, 2025). In DeFi, such scores can support (i) more accurate assessment of on-chain credit risk (e.g., position liquidation or repeated trading losses), (ii) improved risk controls for lending and perpetual trading protocols, and (iii) potential pathways toward more capital-efficient credit (e.g., reduced collateral requirements for low-risk participants) (Packin & Lev-Aretz, 2024).

To avoid ambiguity, the key concepts used throughout this proposal are defined as follows:

- **DeFi (Decentralized Finance):** Financial services (e.g., lending, borrowing, perpetual trading) implemented via smart contracts on public blockchains (Schär, 2021).
- **Smart contract:** Self-executing program code deployed on a blockchain that automatically enforces the terms of an agreement when predefined conditions are met, without a trusted intermediary (Buterin, 2014; Szabo, 1997).
- **Wallet / Address:** A blockchain account that can hold assets and interact with smart contracts (used as the unit of analysis) (Buterin, 2014; Ethereum Improvement Proposals, 2015).

- **On-chain reputation score:** A numerical indicator computed from on-chain signals intended to represent a wallet’s reliability in financial interactions (Do et al., 2023).
- **On-chain credit risk:** Observable adverse outcomes such as position liquidation, repeated trading losses, or protocol-defined delinquency signals on public blockchains (Lin et al., 2024).
- **GMX V2 perpetual swap position:** A collateralized long or short exposure on GMX V2, opened via **PositionIncrease** and closed via **PositionDecrease** events emitted by the GMX V2 **EventEmitter** on Arbitrum One (GMX, n.d.; Arbitrum Foundation, n.d.).
- **Profitable close / close-success:** A **PositionDecrease** with positive realized PnL (**basePnlUsd** > 0) that is not a position liquidation (**orderType** $\neq$ 7) (GMX, n.d.).
- **Close-success count:** The number of profitable **PositionDecrease** events for a wallet within the observation window (analogous to in-degree in transfer-based PageRank validation) (GMX, n.d.; Do et al., 2023).
- **Realized gain proxy:** The sum of $\max(\text{basePnlUsd}, 0)$ over profitable closes in the observation window (analogous to in-value) (GMX, n.d.; Do et al., 2023).
- **Weighted PageRank:** A graph-based ranking method that propagates influence through directed edges weighted by edge strength; this study uses the weighted variant throughout (Page et al., 1998; Do et al., 2023).
- **ERC-20 allowance:** A state variable recording how much a token owner permits a spender to transfer on the owner’s behalf (Ethereum Improvement Proposals, 2015).
- **Approve / Permit:** ERC-20 mechanisms that set allowances; **permit** (EIP-2612) enables allowance updates via signed messages (Ethereum Improvement Proposals, 2020).
- **Endorsement graph:** A directed graph where an edge represents an explicit delegation or endorsement signal rather than a mere transfer (Do et al., 2023).
- **Attestation-based reputation:** A wallet score obtained by aggregating weighted credentials issued about the address holder by third-party verifiers, typically to establish unique person-



hood and resist Sybil attacks (Human Passport, n.d.; Siddarth et al., 2020).

- **Sybil attack:** The creation of many illegitimate identities by a single actor in order to gain disproportionate influence over a pseudonymous system (Siddarth et al., 2020).

Reputation scoring is critical in credit-based and leveraged DeFi to distinguish reliable participants from potential bad actors (Schär, 2021; Packin & Lev-Aretz, 2024). In current practice, trust in a DeFi setting is established in two broad ways, both anchored in what is observable on a public ledger. The first is behavioral: an address accrues standing through consistent, transparent on-chain activity, and because that history is immutable and openly auditable, attempts to inflate it are costly and tend to depress rather than raise the resulting score. The second is attestation-based: tooling such as Human Passport (formerly Gitcoin Passport) binds verifiable credentials to a wallet address and aggregates their weights into a score compared against a passing threshold, deliberately drawing on both on-chain and off-chain evidence in order to counter Sybil attacks (Human Passport, n.d.; Bordeianu & Popescu, 2026). These mechanisms are reviewed in Section 3.4. Their shared objective is to determine whether an address may participate at all; neither is designed to order participating wallets by financial reliability, which is the task addressed here.

Recent PageRank-inspired approaches to on-chain reputation score computation include the AWP algorithm proposed by Do et al. (2023), which assigns reputation scores based on transaction volume (greater inflows imply greater trust) and activeness (recent transactions receive higher weight via a logistic time-decay function). While this approach captures both economic magnitude and temporal relevance, it does not directly model explicit spending authorization as a trust signal.

The ERC-20 token standard, however, includes two explicit allowance mechanisms—approve and permit—by which a token holder grants another address permission to spend up to a specified amount on its behalf (Ethereum Improvement Proposals, 2015; Ethereum Improvement Proposals, 2020). Each approval event inherently represents an endorsement, where the approved amount directly quantifies the granularity of trust. By extracting only the latest allowance for each owner → spender pair, we can build a directed, weighted endorsement graph G_{approve} , in which each edge’s

weight equals the most recent approved token amount (Do et al., 2023). This design eliminates the need for any time-based decay—since newer approvals override older ones—and focuses computation on the most salient, up-to-date trust signals.

Domain validation on GMX V2. This study’s primary empirical setting is **Arbitrum One** and **GMX V2 perpetual swap position trading**. In principle, under-collateralized lending default or liquidation events would provide the most direct credit-risk labels; however, on Arbitrum One the volume of observable, reproducible unsecured or under-collateralized lending activity is insufficient for a stable evaluation dataset at the scale required for this dissertation. We therefore validate EndorseRank against GMX V2 trading-success proxies—long and short perpetual positions closed via **PositionDecrease**—which offer dense, wallet-level realized outcomes on the same chain. Opening a perpetual position is a collateralized leveraged exposure: the trader posts margin and takes directional risk on an underlying asset. Closing the position with positive **basePnLUsd**—a profitable **PositionDecrease**—functionally resembles successfully managing borrowed or leveraged exposure, analogous to borrowing and repaying well in a credit context. Following the AWP paper’s use of in-degree and in-value as domain-intuition proxies, we therefore expect higher reputation scores to align with wallets that achieve more frequent profitable closes (**close-success count**) and larger cumulative realized gains (**realized gain proxy**).

Clarifying the importance in DeFi, enhancing on-chain reputation score computation can provide concrete benefits:

- **Risk management and protocol safety:** Better identification of risky wallets can reduce bad debt and improve the stability of lending and perpetual trading markets (Lin et al., 2024).
- **Capital efficiency:** More informative trust signals can support differentiated risk parameters (e.g., borrowing limits or collateral requirements) instead of one-size-fits-all constraints (Packin & Lev-Aretz, 2024).
- **Access and inclusion without off-chain data:** On-chain scoring can enable participation for users who lack conventional credit records, while preserving the DeFi principle of permis-

sionless access (Schär, 2021).

- **Composable trust for DeFi applications:** A reusable reputation layer can be consumed by multiple protocols (e.g., lenders, perpetual exchanges, insurers, and DAOs) to make consistent, explainable decisions (Packin & Lev-Aretz, 2024).

Despite the clear endorsement semantics and sparser graph structure of approval-based relationships, neither the scholarly reputation models reviewed in Sections 3.1–3.3 nor the attestation-based tooling reviewed in Section 3.4 leverages ERC-20 allowances as primary edge weights for PageRank: the former weight realized transfers, and the latter aggregate credentials about the address holder rather than the authorization relationships between addresses. This methodological gap motivates EndorseRank, our proposed on-chain reputation algorithm that applies a standard Weighted PageRank to G_{approve} . We hypothesize that EndorseRank will achieve comparable or superior alignment with GMX trading-success proxies while significantly reducing data-processing complexity compared to transaction-value– and time-decay–based methods. To ensure a clear alignment between this motivation and what the study will deliver, the proposal specifies objective-linked expected results:

- A reproducible pipeline and latest-allowance endorsement graph derived from ERC-20 approve/permit states on Arbitrum One;
- An optimized, scalable Weighted PageRank implementation (EndorseRank) operating on this sparse, trust-semantic graph; and
- Empirical evidence using GMX V2 **PositionDecrease** outcomes and rank-correlation metrics (Spearman’s rho, Kendall’s tau) showing that EndorseRank matches or improves trading-success alignment relative to transaction-based baselines while reducing computational overhead.

2 Research Problem

Current on-chain reputation methods using PageRank rely primarily on transaction values to infer economic ties among wallets. While the AWP algorithm has shown promise (Do et al., 2023), it exposes two practical limitations when used as a general-purpose on-chain reputation signal. First, reproducing transaction-based PageRank at large chain scale typically requires exhaustive parsing and aggregation of historical transfers, which increases data-processing cost, memory footprint, and iteration time for convergence. This computational burden becomes especially problematic when reputation is intended to be computed repeatedly or near real-time for DeFi risk monitoring on Arbitrum One.

Second, transaction value is an indirect and sometimes ambiguous proxy for “trust.” Wallets may exchange tokens for many reasons (e.g., trading, arbitrage, airdrops, or automated contract activity) without expressing endorsement or delegating spending authority. Yet delegated permission is central to credit exposure in DeFi (Ghosh et al., 2024). By contrast, the ERC-20 allowance mechanism explicitly encodes delegation through approval (and, where supported, permit) updates that authorize a spender to transfer tokens on an owner’s behalf (Ethereum Improvement Proposals, 2015; Ethereum Improvement Proposals, 2020). Because an approval remains in force until changed, the latest allowance state between each owner–spender pair offers a direct, current endorsement signal that can be represented as a sparse directed graph, aligning with network-propagation style risk scoring used in related graph-based models (Lin et al., 2024).

A third consideration concerns the reputation tooling already deployed in practice. Attestation-based systems such as Human Passport establish whether an address is likely to be a distinct person by aggregating weighted on-chain and off-chain credentials against a passing threshold (Human Passport, n.d.; Bordeianu & Popescu, 2026). They therefore act as an admission gate rather than as a graded ordering of wallets, and the relationships between addresses play no part in the score. A delegation-based ranking addresses the complementary question that remains open once such a gate has been applied, and Section 3.4 sets out this distinction in full.

Thus, the central research problem is: Can we construct an on-chain reputation score using ERC-20 approve/permit relationships on Arbitrum One that reduces computational cost, more faithfully represents endorsement semantics, and yields equal or superior alignment with observable GMX V2 perpetual trading success compared to transaction-based PageRank methods?

To address this issue, we will design, implement, and evaluate the EndorseRank algorithm, which constructs a sparse, directed, weighted endorsement graph using ERC-20 approve and permit calls. The key challenges include accurately extracting the most recent approve/permit states between each owner–spender pair so that outdated allowances do not distort the graph; ensuring rapid convergence of the PageRank algorithm on large but sparse endorsement graphs; and evaluating how effectively EndorseRank correlates with GMX V2 trading-success proxies derived from **Position-Decrease** events (close-success count, realized gain proxy, and close success rate).

2.1 Research Questions

Can EndorseRank, a Weighted PageRank computed over ERC-20 allowance relationships, match or outperform transaction-based PageRank baselines in ranking Arbitrum One wallets by trading success, at lower computational cost?

The setting, perspective, comparison baselines, and evaluation criteria that qualify this question are specified in the SPICE framework below rather than restated within the question itself. The sub-questions that follow are grouped into four thematic areas that mirror the evaluation constructs in Section 4.

Table 1: SPICE framework for the main research question

SPICE Element	Description
Setting	Arbitrum One GMX V2 perpetual swap position trading and on-chain reputation score computation

Perspective	DeFi protocols, risk analysts, wallet users, and researchers who need explainable wallet-level risk indicators
Intervention	EndorseRank, a Weighted PageRank metric built from ERC-20 approval/permit-derived allowance relationships
Comparison	Adaptive Weighted PageRank and transfer-based PageRank models that rely on historical transaction flows
Evaluation	Rank divergence, trading-success alignment (Spearman’s rho, Kendall’s tau vs. GMX proxies), runtime, memory use, and convergence stability

The following sub-questions operationalize each SPICE element and are grouped by evaluation theme.

- **Graph Construction and Fidelity**

- RQ1.1: How can we preprocess ERC-20 approve and permit events on Arbitrum One to construct a reliable, up-to-date weighted endorsement graph that reflects current token allowances?
- RQ1.2: Does a graph based on approve/permit edges preserve a sufficient trust signal compared to transaction-value graphs?

- **Ranking Comparison**

- RQ2.1: How does EndorseRank’s ranking output differ from AWP and traditional transaction-based PageRank scoring in terms of relative ordering of wallets?
- RQ2.2: To what extent do EndorseRank scores correlate with AWP scores, and where do they diverge?

- **Trading Success Alignment**

- RQ3.1: What is the statistical correlation (Spearman’s rho, Kendall’s tau) between EndorseRank scores and GMX V2 trading-success proxies (close-success count, realized gain proxy, close success rate)?
- RQ3.2: How does EndorseRank’s alignment with these proxies compare to that of AWP and transfer-based PageRank?

- **Computational Efficiency**

- RQ4.1: What is the execution time and resource usage (CPU, memory) required to compute EndorseRank versus AWP for a fixed set of wallets?
- RQ4.2: How does graph sparsity (approve/permit vs. transfer edges) affect convergence speed and floating-point stability in PageRank iterations?

2.2 Objectives

The overarching goal of this study is to design, implement, and evaluate an EndorseRank algorithm that leverages ERC-20 approve/permit allowances as edge weights without any time decay factors—to produce a computationally efficient and trust-reflective on-chain reputation score. Specifically, we aim to:

The five objective groups below translate this goal into concrete deliverables aligned with the research questions in Section 2.1.

1. Graph Construction

- **O1.1:** Build a data-extraction pipeline that scans **Arbitrum One** for ERC-20 **Approve** and **Permit** events, isolates the most recent allowance amount for each owner → spender pair, and stores these values.
 - *Rationale:* Since each new approve/permit transaction overrides previous allowances for the same owner → spender, only the latest allowance per pair needs to be retained.
- **O1.2:** Construct a directed, weighted graph G_{approve} in which each node represents an ERC-20

wallet, and each edge ($u \rightarrow v$) carries a weight equal to the **latest allowance amount** (e.g., in ETH or token units) from wallet u (owner) to wallet v (spender).

- *Detail:* We do **not** apply any time-based decay or history aggregation; all older approvals are superseded by the most recent event.

2. EndorseRank Algorithm Implementation

- **O2.1:** Implement a **Weighted PageRank** algorithm on G_{approve} , where each node’s “score” is computed by distributing its PageRank mass to outgoing edges proportionally to the allowance weights.

- *Specifications:*

- (a) Normalize each node’s outgoing edges so that for node u , the probability of transitioning to neighbor v is:

$$P_{u \rightarrow v} = \frac{w_{u \rightarrow v}}{\sum_{x \in N^+(u)} w_{u \rightarrow x}}$$

where $N^+(u)$ is the set of out-neighbors of u (all nodes x that u approves), x is an element of $N^+(u)$, and $W_{u \rightarrow v}$ is the latest allowance amount from wallet u to wallet v .

- (b) The equation above specifies the transition probability used by EndorseRank: $P_{u \rightarrow v}$ equals $W_{u \rightarrow v}$ divided by the sum of all outgoing allowance amounts from wallet u , where each $W_{u \rightarrow v}$ denotes the latest allowance amount on edge $u \rightarrow v$. Thus, each wallet distributes its PageRank mass only across currently active approval relationships, proportionally to normalized allowance strength.
 - (c) Use a standard damping factor (e.g., 0.85), but do not incorporate any additional time-weighting or value-decay beyond the raw allowance. This keeps the transition model consistent with PageRank-style convergence guarantees, avoids tuning extra hyperparameters without empirical justification, and preserves comparability to baselines that are defined over raw allowance weights rather than time-adjusted scores.
- **O2.2:** Optimize the EndorseRank implementation for large, sparse graphs by using sparse-

matrix representations and efficient iterative solvers, ensuring that convergence is achieved with minimal memory overhead and fast runtime.

3. Reproduction of Baselines and Data Consistency

- **O3.1:** Reproduce two baseline ranking approaches on an identical wallet set drawn from the **GMX V2—active trader population on Arbitrum One** over the same time window:
 - (a) Adaptive Weighted PageRank (AWP): a transaction-value-weighted PageRank that applies time-decay factors to transfers.
 - (b) Transfer-Based Weighted PageRank: a simpler model that uses raw ERC-20 transfer amounts as edge weights, without time decay.
- **O3.2:** Maintain a uniform data-processing pipeline so that EndorseRank, AWP, and Transfer-Based PageRank all operate on the same set of wallets and the same period’s events, ensuring that any differences in results are due solely to the ranking logic rather than data inconsistencies.

4. Performance Evaluation

- **O4.1:** Compute Spearman’s rho and Kendall’s tau between EndorseRank scores and each of the two baselines (AWP and Transfer-Based PageRank) to quantify how similarly or differently wallets are ordered. Spearman’s rho captures monotonic rank agreement, while Kendall’s tau measures pairwise ordering agreement.
- **O4.2:** Compare the distribution of scores (e.g., mean, median, interquartile range) for EndorseRank versus the baselines, identifying any outlier wallets that receive disproportionately high approval-based weights but low transfer-based weights (or vice versa).
- **O4.3: Trading Success Alignment**
 - Extract GMX V2 **PositionDecrease** events from the GMX V2 **EventEmitter** on Arbitrum One, identifying traders via the event **account** field (not **tx.from**).
 - Compute trading-success proxies for each wallet: **close-success count** (profitable closes),

realized gain proxy ($\sum \max(\text{basePnlUsd}, 0)$), and **close success rate** (wins / total closes, with $\text{total_closes} \geq 3$).

- Compute Spearman’s rho and Kendall’s tau between EndorseRank, AWP, and transfer-based PageRank scores and each proxy ranking, following the domain-alignment approach used in Do et al. (2023) for in-degree and in-value.
- Interpret correlation magnitudes: values too close to 1 indicate redundancy with simple proxy rankings; values below approximately 0.2 suggest insufficient domain insight from the PageRank propagation.

5. Computational Efficiency and Scalability Analysis

- **O5.1 Runtime Benchmarking:** Measure and compare the total time required (data extraction + graph construction + PageRank convergence) for EndorseRank, AWP, and Transfer-Based PageRank on a fixed wallet set, with staged expansion to 100,000 nodes (10,000 pilot → 50,000 → 100,000) after dry-run cost checks, subsampled from locally cached parquet graphs.
- **O5.2 Memory Profiling:** Record peak memory usage for each method, highlighting how EndorseRank’s use of only the latest approval amounts results in a sparser graph and lower memory footprint than transaction-based graphs.
- **O5.3 Convergence Stability:** Analyze the number of iterations needed for each algorithm to reach a predefined convergence tolerance (e.g., 10^{-6}) under different damping factors and token supply scales, verifying that EndorseRank remains numerically stable despite potentially large allowance values.

SMART Alignment

The objectives are specific because each one targets a concrete research output: data pipeline, graph construction, algorithm implementation, baseline reproduction, trading-success alignment evaluation, and scalability benchmarking. They are measurable through rank-correlation statistics (Spearman’s rho, Kendall’s tau), runtime, memory, and convergence tolerance. They are attainable and realistic because they use public Arbitrum One blockchain logs, GMX V2 **PositionDe-**

crease events, reproducible Python-based tooling, and fixed computational resources. They are time-bound through the post-approval schedule, which allocates data extraction, implementation, evaluation, writing, and publication milestones between September 2026 and January 2028.

By accomplishing these objectives, we will demonstrate how a latest-allowance-weighted approach to PageRank can yield a scalable, trust-aligned reputation-scoring model, EndorseRank, that outperforms or matches existing transaction-value-based methods in GMX trading-success alignment without any time-decay overhead.

2.3 Expected Results

This section states the expected outcomes of the study and explicitly aligns them with the objectives above.

These expected results map directly onto Objectives O1-O5 and mirror the three themes introduced in the background: endorsement-based trust structure, GMX trading-success alignment, and computational efficiency relative to transfer-based baselines.

ER1: Reproducible allowance dataset and endorsement graph (O1.1-O1.2)

A documented, reproducible data-extraction pipeline that outputs the latest ERC-20 approve/permit allowance state for each owner-spender pair over a defined time window on Arbitrum One.

A directed, weighted endorsement graph G_{approve} constructed from these latest-allowance states, accompanied by summary statistics (e.g., node/edge counts, sparsity, degree distributions) that substantiate the intended sparsity relative to transfer graphs.

ER2: Scalable EndorseRank implementation (O2.1–O2.2)

A correct Weighted PageRank implementation that uses latest allowance amounts as edge weights (with a clearly specified damping factor and convergence tolerance) and is optimized for large, sparse graphs.

Evidence of numerical stability under large allowance values and practical convergence behavior

(iterations to tolerance) on graphs of increasing size.

ER3: Baseline reproduction and ranking-difference evidence (O3.1-O4.2)

Reproduced baseline rankings (AWP and transfer-based weighted PageRank) computed on the same wallet set and the same observation window as EndorseRank.

Quantified agreement and divergence in wallet ordering via Spearman's ρ^* and Kendall's τ^* , alongside targeted case analyses for wallets that are highly ranked by allowance endorsement but not by transfer volume (and vice versa).

ER4: Trading-success alignment results (O4.3)

A GMX V2 **PositionDecrease** dataset with computed close-success count, realized gain proxy, and close success rate for matched wallets, and comparative rank-correlation results showing whether EndorseRank yields comparable or superior alignment with these proxies relative to AWP and transfer-based PageRank (Spearman's rho and Kendall's tau).

ER5: End-to-end efficiency and scalability benchmarks (O5.1-O5.3)

Runtime and memory profiling results for (data extraction + graph construction + PageRank convergence) demonstrating the computational advantage of using only the latest allowance state versus exhaustive transfer-history parsing.

A scalability analysis that reports how graph sparsity affects convergence speed and resource consumption across methods.

3 Literature Review

3.1 Graph-Based On-Chain Reputation Scoring

Although prior work often uses Ethereum mainnet as the reference chain, **this dissertation’s primary empirical setting is Arbitrum One with GMX V2 perpetual swap position outcomes** for domain validation. The literature below remains relevant as background for graph-based reputation methods.

Early research on on-chain reputation leveraged graph theory, particularly adaptations of PageRank, to rank wallet addresses by “trustworthiness” (Page et al., 1998). Do et al. (2023) model Ethereum addresses as nodes in a directed graph with transactions as edges, then apply the PageRank algorithm to compute a social credit score for each address. The rationale is that DeFi’s growth demands methods to identify and rank entities for unsecured lending and customer protection. In this framework, an address accumulates higher score if it is “trusted” by many or large transactions from others, analogous to how important web pages have many incoming links. PageRank thus measures connectivity strength and activity as proxies for reputation. Subsequent graph models combine multiple factors (e.g., frequency and volume of interactions) into edge weights (Do et al., 2023; Vathsala V. & V. D. Sharma, 2025).

A key consideration in graph construction is how to weight the edges. Basic implementations treat each transaction equally, but later work incorporates transaction value as a weight, so that large transfers signify stronger trust. Using a HodgeRank approach, Do et al. (2023) treat the amount of ETH transferred as the weight on each edge and then solve for a global ranking that accounts for these weighted flows. This produces an Ethereum reputation ranking influenced by transaction volume, under the intuition that high-value interactions indicate greater trust or stake between parties. Another practical challenge is dealing with graph structure: Ethereum’s transaction graph can contain many isolated clusters or “sink” nodes that absorb PageRank score. Damping factors, as in traditional PageRank, are used to mitigate score inflation for sinks and small isolated subgraphs,

ensuring the reputation scores reflect participation in the broader network rather than siloed activity. One limitation of a vanilla PageRank, i.e., the unmodified PageRank formulation applied to a transaction graph without explicit risk propagation or domain-specific adjustments, is that it may not distinguish malicious but active actors from genuinely reputable ones. Lin et al. (2024) show that transaction connectivity alone can miss risk-relevant structure unless risk signals are propagated through the network. This highlights the need for refined models. Proposed improvements include incorporating negative feedback or propagation of risk labels through the network. For instance, recent research introduced RiskProp, a network-propagation-based risk scoring model for Ethereum accounts (Lin et al., 2024). RiskProp augments the transaction graph with a "de-anonymous score" for addresses and propagates risk signals through directed bipartite graphs. The approach was able to flag a majority of known high-risk accounts by propagating risk indicators across transaction links, effectively identifying suspicious wallets that a simple connectivity measure might overlook. This illustrates how graph-based methods can be extended beyond pure PageRank to serve as on-chain analogues of credit risk models, where the "credit" in question may be financial reliability or compliance risk.

It is worth noting that graph-based reputation systems often consider temporal dynamics. Real-world creditworthiness changes over time, so some models incorporate time-decay factors to diminish the influence of stale interactions. Do et al. (2023) apply logistic time-decay to transfers in AWP so that recent activity contributes more strongly than stale edges, while Lin et al. (2024) show that risk-relevant structure can be masked unless propagation models are updated as network state evolves. A dynamic reputation model might weight recent transactions more heavily or periodically expire old edges. This prevents entities from resting on long-past positive interactions and addresses the whitewashing problem, where an entity's bad behavior gets diluted by time. However, striking the right balance is crucial: too much decay can ignore longstanding trust relationships, while too little can allow outdated information to skew the scores.

3.2 Predictive Modeling for On-Chain Credit Risk

In parallel with graph-theoretical methods, researchers have developed data-driven machine-learning models for on-chain credit scoring, especially in DeFi lending contexts. These models take inspiration from traditional credit analytics, using wallet histories as input features and predicting the likelihood of default or delinquency. Ghosh et al. (2024), in an arXiv preprint, propose an on-chain credit risk scoring framework for decentralized finance that featurizes wallet activity from public event logs and trains supervised models to rank borrowers by repayment and liquidation risk. In their framework, each wallet’s activity is featurized analogously to a credit report: features include payment history, amounts owed, length of credit history, new credit, and credit mix, all derived from on-chain behavior. Using machine-learning classifiers, the model estimates the probability that a given on-chain lending position will become delinquent (e.g., fall below collateral requirements and face liquidation) in the near future. The output is essentially a credit risk score for the wallet or position, indicating the odds of default. Ghosh et al. report that this approach can successfully rank borrowers by risk and thus support under-collateralized lending by relying on predicted probability of repayment rather than collateral ratios alone.

Other machine-learning approaches similarly treat on-chain transactions and account data as inputs to predictive models for creditworthiness. Hassija et al. (2020) presented a blockchain-based credit scoring model that leverages prospect theory to weight positive vs. negative credit events differently, capturing the asymmetric risk attitudes of borrowers. Their model, implemented as a smart contract, calculates a trust score for each participant which is updated as new lending outcomes occur.

By incorporating prospect theory (which, for instance, might penalize defaults more heavily than it rewards timely repayments), the scoring mechanism aligns with human risk biases, potentially improving prediction of who will default. Similarly, Chakraborty et al. (2019) propose a blockchain-based credit analysis framework that stores and evaluates customer credit data on-chain for more efficient financial-system workflows. Their design demonstrates how distributed ledger structures

can support auditable credit assessment alongside conventional transaction histories.

3.3 Industry Initiatives and Scholarly Perspectives on DeFi Credit Scoring

Industry demand for on-chain credit scoring has prompted both commercial deployments and scholarly analysis of decentralized lending workflows. Molnár et al. (2023) examine blockchain-based business process management in finance and show how credit and claim requests can be re-engineered through smart-contract workflows, highlighting the need for transparent, auditable risk signals in DeFi lending. Packin and Lev-Aretz (2024) analyze hybrid on-chain/off-chain credit architectures and the governance challenges of integrating traditional credit data with blockchain-native activity, underscoring that reputation metrics must remain interpretable under pseudonymity constraints. Recent scholarly work also treats DeFi credit scoring as a measurable research problem rather than a purely commercial feature. Shimpi (2024) implements an Ethereum-based lending workflow that integrates traditional credit-scoring components with smart-contract credit bureau functions, illustrating how off-chain credit data can be bridged to on-chain lending decisions. Vathsala V. and V. D. Sharma (2025) propose a secure AI architecture for dynamic DeFi credit scoring that combines federated learning with blockchain-anchored data provenance, noting that most deployed systems rely on transfer histories or loan outcomes rather than explicit authorization relationships. These scholarly perspectives highlight current gaps that EndorseRank aims to fill. Existing implementations focus heavily on past transaction behavior (payments, liquidations, balances) or supervised default prediction, but they do not systematically leverage ERC-20 token approval relationships between wallets as primary graph edges. An ERC-20 approve (or permit) action, which authorizes another address to spend tokens on the owner’s behalf (Ethereum Improvement Proposals, 2015; Ethereum Improvement Proposals, 2020), is a strong signal of trust or financial linkage that lies outside direct token transfers. Current graph-based scoring systems largely ignore these approval edges, focusing only on realized transfers of value. EndorseRank addresses this research gap by integrating ERC-20 approve/permit edges as first-class links in the reputation graph, treat-

ing them as directed weighted edges in a PageRank computation. **Because trust in operational**  DeFi systems is established not only through the scholarly models reviewed so far but also through widely deployed identity and reputation tooling, Section 3.4 examines that practice explicitly and states how EndorseRank differs from it.

Furthermore, EndorseRank is distinctive in its handling of temporal effects. Whereas many reputation systems apply time-based decay to down-weight old interactions, EndorseRank proposes to forgo time decay for approval/permit edges. The rationale is that an approval remains in force until revoked; a permission granted a year ago is equally pertinent today if the allowance is still active. By giving persistent weight to long-lived trust links, EndorseRank preserves important information about enduring financial relationships that a short memory window would lose. This contrasts with typical transaction-based scoring, where a payment from years ago might reasonably be given little importance. In the context of approvals, the lack of time decay reflects the enduring nature of the risk: as long as B is authorized to spend A's funds, A's exposure (and hence the reputational linkage) does not diminish over time. This no-decay approach is distinctive, as most prior works have either not included such edges at all or have not considered the temporal aspect of trust explicitly. By integrating ERC-20 approval networks with PageRank and maintaining their influence over time, EndorseRank is expected to provide a richer and more stable on-chain reputation score—one that captures both the flow of value and the web of delegated permissions on Arbitrum One.

3.4 Attestation-Based Reputation Tooling in Deployed DeFi Systems

Alongside the scholarly models reviewed above, wallet-level trust in operational DeFi and web3 systems is most commonly established through identity and reputation tooling bound directly to a wallet address. The most widely deployed instance is Human Passport (formerly Gitcoin Passport), which aggregates verifiable credentials—termed Stamps—issued for government-identity (KYC) checks, biometrics, web-of-trust vouching, and prior web2 and web3 activity. Each Stamp carries a weight that reflects its cost of forgery, a wallet's score is the weighted sum of the Stamps it holds,

and that score is compared against a passing threshold used to admit or exclude the address (Human Passport, n.d.). Stamps are verified off-chain and may optionally be minted on-chain as attestations through the Ethereum Attestation Service, so that smart contracts can read a wallet’s score without relying on an external server (Human Passport, n.d.).

Two properties of this design bear directly on the present study. First, the score deliberately combines on-chain and off-chain evidence, because its purpose is Sybil resistance: establishing that an address is controlled by a distinct person rather than being one of many addresses operated by a single actor. Second, the score functions largely as a gate—a wallet either meets the threshold or it does not—rather than as a graded ordering of wallets by financial reliability. Comparative security analysis classifies Passport within the attestation-stacking family of Sybil defences, reports strong but not decisive discrimination relative to biometric and social-graph alternatives, and identifies credential farming, in which an adversary acquires low-cost Stamps in bulk, as its principal residual weakness (Bordeianu & Popescu, 2026). The tooling originates in quadratic-funding mechanisms, whose matching allocations rest on a one-contributor-one-voice assumption and are therefore acutely sensitive to duplicated identities (Buterin et al., 2019).

This practice reflects a broader line of work on proof of personhood. Siddarth et al. (2020) review protocols that establish Sybil-resistant identity through subjective inputs such as voting, vouching, and interpretation, and observe that such systems must trade off Sybil resistance, self-sovereignty, and privacy against one another. Whatever substrate is used, the question these protocols answer is whether an identifier corresponds to a unique person, not how one wallet should be ranked against another for credit-like purposes.

Passport additionally offers model-based scoring that classifies any EVM address as human or Sybil in real time from its on-chain activity across Ethereum and several Layer-2 networks, including Arbitrum, by comparing that activity against curated lists of known Sybil and human addresses (Human Passport, n.d.). This confirms that purely on-chain behavior carries usable trust signal on the same chain studied here. The features involved, however, are per-address activity aggregates evaluated against a binary human-or-Sybil target; the directed relationships between wallets are not

themselves the object of measurement.

EndorseRank is therefore neither a competitor to nor a replacement of attestation-based tooling, and the difference is best stated along five dimensions:

- **Signal source:** Passport aggregates third-party attestations about the person behind an address; EndorseRank derives its edges from a financial act that wallets themselves record on-chain, namely an ERC-20 approval authorizing a spender to move the owner's tokens.
- **Data scope:** Passport combines off-chain credentials (KYC, biometrics, web2 accounts) with on-chain activity; EndorseRank uses public on-chain state only and requires no personal data, preserving permissionless participation.
- **Question answered:** Passport asks whether an address is a distinct human; EndorseRank asks how a wallet ranks relative to other wallets within a network of delegated spending authority.
- **Output form:** Passport produces a score interpreted against a pass threshold; EndorseRank produces a continuous, network-propagated ranking suitable for differentiated risk parameters.
- **Validation target:** Passport-style scores are assessed against Sybil-labelled address sets in grant and airdrop distribution; EndorseRank is assessed against realized GMX V2 perpetual trading outcomes on Arbitrum One.

The two are complementary rather than mutually exclusive: a personhood gate can remove duplicated identities before a relational reputation score is computed over the remaining population, and Section 6 treats the absence of such a gate in this study as an explicit limitation. What neither the scholarly literature of Sections 3.1–3.3 nor the deployed tooling described here has examined is whether the delegation relationships already recorded in ERC-20 allowances constitute a usable trust signal in their own right. That is the specific gap this study addresses, and it is the basis on which the study claims an original contribution.

In summary, the literature shows a progression from simple graph-based reputation metrics to machine-learning models for on-chain credit scoring, running in parallel with an industry prac-

tice that scores wallets through aggregated identity attestations. PageRank-based techniques contribute a transparent, explainable, network-centric view of trust and influence; machine-learning models offer risk-ranking performance using diverse features; attestation-based tooling supplies personhood assurance but not a graded ordering of financial reliability. A gap remains between them: incorporating meaningful relational graph features, such as ERC-20 approval links, into a scoring system that is at once explainable, permissionless, and domain-aligned. EndorseRank aims to bridge that gap. It builds on the insight that network structure matters—who authorizes whom—and that this structure can be quantified through a PageRank-style algorithm. Empirical validation in this dissertation uses GMX V2 perpetual swap position outcomes on Arbitrum One: close-success count and realized gain proxy serve as observable trading-success measures, analogous to the in-degree and in-value proxies used in Do et al. (2023). By doing so without arbitrarily discarding older trust links, EndorseRank preserves the long-term context of relationships. This approach addresses limitations in prior art and aligns with scholarly calls for transparent, composable DeFi risk assessment (Molnár et al., 2023; Shimpi, 2024). It promises an enhanced on-chain reputation score that could improve risk assessment for leveraged DeFi activity on Arbitrum One, making decentralized finance both more accessible and more secure.

4 Theoretical Framework

This study is grounded in hyperlink-citation theory, construct (domain) alignment, and computational complexity theory to compare EndorseRank against Adaptive Weighted PageRank (AWP) across three core constructs: Reputation Structure, Trading Success Alignment, and Computational Efficiency. Each construct links directly to the parameter definitions introduced earlier in the research objectives and expected-results framework and supports testable hypotheses.

4.1 Constructs and Variable Mapping

Table 1 maps each evaluation construct to its variables and operational measures.

Table 2: Evaluation Constructs and Associated Variables

Construct	Associated variables	Operational measures
Reputation Structure	EndorseRank score, AWP score, rank ordering, rank divergence	Spearman’s rho, Kendall’s tau, top-k overlap
Trading Success Alignment	GMX close-success count, realized gain proxy, close success rate, reputation scores	Spearman’s rho, Kendall’s tau vs. each proxy
Computational Efficiency	Graph build time, PageRank convergence time, memory usage, iteration count	Runtime, peak memory, iterations to tolerance

4.2 Conceptual Model

Figure 1 presents the conceptual model: the EndorseRank vs. AWP comparison is evaluated across three constructs (Reputation Structure, Trading Success Alignment, and Computational Efficiency),

each with a corresponding testable hypothesis (H1–H3).

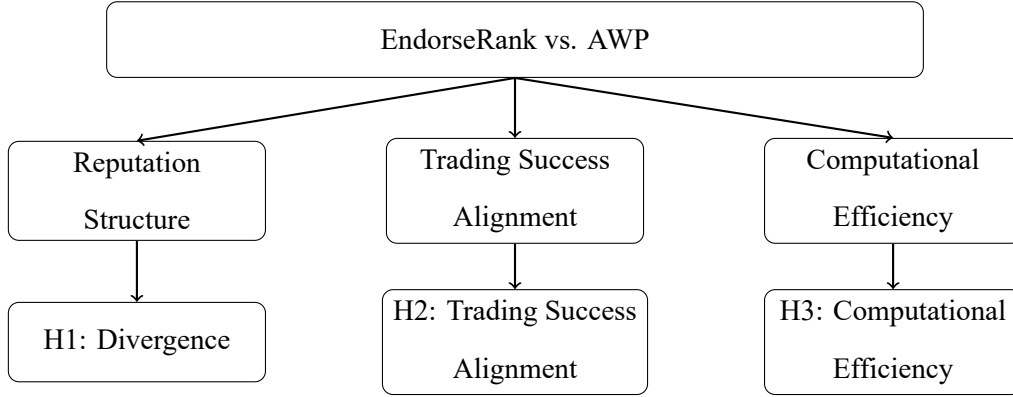


Figure 1: EndorseRank vs. AWP: hypotheses and evaluation constructs

H1 (Divergence): EndorseRank and AWP will yield significantly different wallet orderings, demonstrating that an approve/permit-based endorsement graph creates a distinct reputation structure.

H2 (Trading Success Alignment): EndorseRank’s scores will correlate more strongly with GMX V2 trading-success proxies (close-success count and realized gain proxy) than AWP’s scores, indicating superior domain-intuition alignment.

H3 (Computational Efficiency): EndorseRank will require less runtime, memory, and fewer iterations to converge than AWP, demonstrating practical scalability advantages.

4.3 Theoretical Justification

Hyperlink–Citation Semantics: The original PageRank algorithm (Page et al., 1998) evaluates a webpage’s importance by the structure of inbound hyperlinks, treating each link as a citation and modeling a random surfer who follows links with a damping probability. Do et al. (2023) apply this propagation logic to Ethereum transaction graphs, weighting edges by transfer value and time-decay so that reputation flows along observed economic ties. This study extends the same citation-based propagation to ERC-20 approve/permit edges because allowances are explicit, on-chain delegations of spending authority (Ethereum Improvement Proposals, 2015; Ethereum Improvement Proposals, 2020). Unlike token transfers, which may reflect trading or automated routing without

trust intent, an allowance records a durable authorization state until revoked. Graph-based risk models on Ethereum (Lin et al., 2024) further show that network structure carries risk-relevant information beyond raw activity counts, supporting the use of a directed endorsement graph rather than transfer volume alone.

Domain Alignment Framework: In measurement theory, construct alignment assesses how well an indicator tracks an observable outcome of interest (Cronbach & Meehl, 1955). For rank-based reputation scoring, Spearman’s rho and Kendall’s tau are appropriate because they evaluate monotonic and pairwise ordering agreement between reputation scores and domain proxies without assuming a classification threshold (Do et al., 2023). Following Do et al. (2023), who validated PageRank against in-degree and in-value, we treat GMX V2 **close-success count** and **realized gain proxy** as observable trading-success measures on Arbitrum One. Reputation scores that align with these proxies therefore satisfy the study’s domain-intuition requirement without claiming external ground-truth loan-repayment labels.

Leveraged Exposure and Trading Success Alignment: GMX V2 perpetual swap positions represent collateralized leveraged exposure: the trader posts margin and takes directional risk on an underlying asset through long or short positions. A profitable **PositionDecrease** close (positive **basePnlUsd**, excluding position liquidation) functionally resembles successfully managing borrowed or leveraged exposure—analogous to borrowing and fulfilling obligations in a credit context.

Computational Complexity Theory: PageRank’s time complexity per iteration is $O(|E|)$ (Page et al., 1998). Approve/permit graphs are sparser than transfer-based graphs (fewer edges $|E|$), leading to lower per-iteration cost, faster convergence, and reduced memory usage.

Scope and Future Extensions: This proposal prioritizes interpretable graph propagation with verifiable on-chain signals over black-box predictive models. Supervised and representation-learning approaches to DeFi credit scoring (Ghosh et al., 2024; Kotb, 2023) can improve ranking performance but often sacrifice transparency and auditability under pseudonymity constraints emphasized by Packin and Lev-Aretz (2024). EndorseRank therefore stays within the PageRank family so that

each score remains traceable to explicit allowance edges. Future work may combine endorsement-graph features with DeFi-informed representation learning, but such extensions are outside the present study’s scope.

4.4 Implications for Research Design

The research design applies the constructs above through the following measurement and analysis steps.

Measurement: Compute EndorseRank and AWP scores for a matched sample of GMX V2–active wallets on Arbitrum One, derive close-success count, realized gain proxy, and close success rate from **PositionDecrease** events over a defined observation window, and profile each algorithm’s runtime, memory usage, and iterations to convergence.

Hypothesis testing and interpretation follow the measurement steps above and are organized as follows.

Analysis:

H1 via rank agreement between EndorseRank and AWP scores (Spearman’s ρ , Kendall’s τ , top- k overlap), testing whether these coefficients differ significantly from perfect concordance

H2 via rank correlation between reputation scores and GMX trading-success proxy rankings (Spearman’s ρ , Kendall’s τ). Spearman’s rho captures monotonic rank agreement; Kendall’s tau measures pairwise ordering agreement. These metrics are justified because reputation scores and trading-success proxies are continuous rank variables rather than binary default labels, following the domain-alignment approach in Do et al. (2023).

H3 via paired benchmarks of runtime, peak memory, and iteration counts

Contribution: By anchoring our hypotheses in established theories, this framework ensures that empirical results will offer both scholarly rigor and practical relevance, demonstrating EndorseRank’s advantages in producing a semantically meaningful, domain-aligned, and efficient on-chain reputation metric.

5 Research Methodology

This section outlines the overall approach for implementing EndorseRank, comparing it to Adaptive Weighted PageRank (AWP), and evaluating their performance in terms of reputation structure, trading success alignment, and computational efficiency.

5.1 Research Design and Method

Dataset Overview

This study uses public, on-chain event logs as the primary dataset. The raw data are time-stamped, address-level records emitted by Arbitrum One smart contracts, which makes them reproducible but also inherently pseudonymous (wallets are identifiers, not verified real-world entities).

Core unit of analysis

Wallet addresses and directed (one-way) relationships from owner to spender, later represented as a sparse graph. GMX V2 trading outcomes are aggregated at the same wallet level using the **account** field from **PositionDecrease** events.

Primary signals

ERC-20 Approve (and EIP-2612 Permit) authorizations that specify an owner, a spender, and an allowance amount (with token contracts providing the context for each authorization). GMX V2 **PositionIncrease** and **PositionDecrease** events provide realized trading outcomes for validation.

Sparsity and heavy tails

Endorsement/authorization graphs are expected to be highly sparse with skewed degree/weight distributions (a small number of contracts/routers receive many approvals), which motivates robust evaluation beyond mean-based summaries.

Temporal structure

Events form a time series; we operationalize EndorseRank on a fixed observation window (December 1, 2025 – May 31, 2026) and construct “latest state” edge weights by retaining the most recent relevant allowance information per relationship. GMX trading-success proxies are computed over

the same window.

Data quality considerations

Event logs are generally reliable but can include revocations (`amount = 0`), “spend-all by default” approvals, and occasional indexing gaps; these characteristics are treated explicitly in preprocessing and robustness checks. GMX events must be decoded from the EventEmitter **EventLog1** format; trader identity uses the event **account** field, not `transaction.from`, to avoid misattributing router or keeper addresses.

Data Collection & Preprocessing

Blockchain Data Source

Arbitrum One (chain ID 42161), observation window December 1, 2025 – May 31, 2026, via Google BigQuery (`bigquery-public-data.google_blockchain_arbitrum_one_us`) or an archive node.

Event Extraction

Approval/permit-derived authorizations: all ERC-20 Approval(owner, spender, value) events on Arbitrum One, including approvals generated through permit-compatible flows where the token emits the standard Approval event.

GMX V2 perpetual swap outcomes: **PositionDecrease** logs from the GMX V2 **EventEmitter** contract (`0xC8ee91A54287DB53897056e12D9819156D3822Fb`) over the full observation window are the **primary validation stream**, decoded to extract **account**, **basePnlUsd**, **orderType**, and liquidation flags. **PositionIncrease** events are referenced for domain context (position opening) but are not extracted separately. Profitable closes satisfy **basePnlUsd** > 0 and are not position liquidations (`orderType` $\neq 7$).

Empirical Validation Choice

Theoretically, the strongest validation target for on-chain credit reputation would be a binary default or liquidation label drawn from under-collateralized lending (e.g., loan liquidation after health-factor breach). Such labels directly encode credit failure on-chain. In practice, however, the Arbi-

trum One ecosystem targeted by this study lacks sufficient active, publicly observable unsecured or under-collateralized lending volume to construct a large, reproducible wallet-level default dataset within the observation window. Collecting comparable labels from other chains would break the single-chain design and inflate data-engineering cost without improving endorsement-graph semantics.

Accordingly, this dissertation adopts **GMX V2 perpetual long/short position outcomes** as the primary empirical validation frame. Collateralized perpetual trading on GMX produces high-frequency, wallet-attributed **PositionDecrease** events with realized PnL, enabling measurable proxies—close-success count, realized gain proxy, and close success rate—for credit-like risk management on the same chain as the endorsement graph. These measures are interpreted as **domain-intuition proxies**, not ground-truth loan-repayment labels; limitations are discussed in Section 6.

Trading-Success Proxies (Validation Targets)

Following Do et al. (2023), who validated PageRank against in-degree and in-value, we define GMX analogs:

- **Close-success count:** number of profitable **PositionDecrease** events per wallet in the observation window.
- **Realized gain proxy:** $\sum \max(\text{basePnlUsd}, 0)$ over profitable closes in the window.
- **Close success rate:** wins / total closes, retaining wallets with $\text{total_closes} \geq 3$ for statistical stability.

Sampling

The alignment analysis uses the intersection of wallets present in the reputation graph and wallets with at least three GMX **PositionDecrease** events in the observation window. Computational benchmarks use a reproducible random sample with a fixed seed, expanded in stages (10,000 → 50,000 → 100,000 wallets) after dry-run cost checks; robustness checks compare 50,000- and 200,000-wallet subsamples drawn from locally cached parquet graphs (no additional BigQuery scans).

BigQuery Data Acquisition Plan

Data are collected in three phases from `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`. Technical details and SQL templates are versioned in `A-Skill-Programs/margin_rank/docs/bigquery_data_plan.md`.

Phase 1 — GMX validation labels (Stream A).

Extract all GMX V2 **PositionDecrease** logs from the EventEmitter contract over the full six-month observation window. Decode **account**, **basePnlUsd**, and **orderType**; identify qualified wallets (≥ 3 closes). This phase is highly selective (contract + topic filters) and is expected to scan on the order of 1–10 GiB.

Phase 2 — Reputation subgraph (Streams B & C).

For the Phase 1 wallet set (plus benchmark seed addresses), extract ERC-20 **Approval** and **Transfer** events in six monthly batches. Preprocess to latest nonzero allowances (EndorseRank) and time-decayed transfer edges (AWP). Wallet filters reduce returned rows; partition scans remain bounded by monthly time windows and event-topic filters.

Phase 3 — Evaluation and benchmark (local).

Run alignment and rank-divergence tests on Tier 1 parquet outputs. Scale computational benchmarks to 10k, 50k, and 100k wallets by subsampling the cached subgraph—no additional BigQuery queries.

Tier	Cohort / scale	Est. scan	Hypotheses
Tier 1 (required)	GMX ≥ 3 closes; all qualified wallets (expected 500–3,000)	~80–350 GiB	H1, H2
Tier 2 (staged)	Benchmark pilot 10k $\rightarrow$ mid 50k $\rightarrow$ full 100k (parquet subsample)	≈ 0 extra BQ	H3
Tier 3 (local)	Damping, top-token subgraph, weight transforms	0 (local recompute)	Robustness

Cost guardrails.

All extractions require a BigQuery `--dry-run` before `--extract`; per-query scan is capped at 150 GiB (`margin_config.yaml`). The billing account receives 1 TiB of query processing free per month; overage is approximately US\$6.25/TiB. Target total cost for Tier 1+2 is **\$0**. If dry-run estimates exceed budget, restrict Phase 2 to the top ERC-20 tokens by Arbitrum volume (consistent with Section 5.2.2 robustness checks). Bytes processed and row counts are logged in `data/processed/extraction_manifest.json`.

Construct	Definition	Associated Variables
Reputation Structure	The pattern and relative ordering of wallet reputations produced by a ranking algorithm.	(1) EndorseRank score; AWP score; Rank correlation (Spearman's ρ / Kendall's τ) between EndorseRank and AWP
Trading Success Alignment	The extent to which a reputation score aligns with GMX V2 trading-success proxy rankings.	Close-success count; Realized gain proxy; Close success rate; Spearman's ρ / Kendall's τ for each ranking method vs. each proxy
Computational Efficiency	The computational resources(time, memory, convergence iterations) required to compute scores.	EndorseRank runtime; AWP runtime; Peak memory usage; Iterations to convergence

1. Graph Construction & Algorithm Implementation

- **Approve/Permit Graph** (G_{approve})

- **Nodes:** Wallet addresses.
- **Edges:** directed owner-to-spender relationships weighted by the latest nonzero allowance from u to v . Because ERC-20 tokens differ in decimals and economic scale, weights will first be token-decimal adjusted; sensitivity checks will compare raw allowance weights, log-transformed weights, and value-normalized weights where reliable price data are available.

- **Transfer-Based Graph** (G_{transfer})

- Same node set; edges weighted by sum of transfer amounts $u \rightarrow v$.

- Algorithms

EndorseRank: Weighted PageRank on G_{approve} with damping factor $d = 0.85$. This value is retained because 0.85 is the standard PageRank setting, provides a stable balance between diffusion and teleportation, and enables direct comparison with prior graph-ranking studies.

- **AWP Reproduction:** Adaptive Weighted PageRank on G_{transfer}

incorporating time and value decay per Do et al. (2023).

- Tools & Environment

- Python with NetworkX or GraphFrames; Scipy sparse solvers; Docker containers for reproducibility.

2. Performance Evaluation Procedures

- Trading Success Alignment

- For each ranking method (EndorseRank, AWP, transfer-based PageRank), compute Spearman’s ρ and Kendall’s τ against close-success count, realized gain proxy, and close success rate rankings on the matched wallet sample.

- Compare EndorseRank vs. AWP alignment coefficients; test whether EndorseRank achieves higher correlation without approaching 1.0 (which would indicate redundancy with simple proxy rankings).
- Report results in tables analogous to Do et al. (2023), including comparison against a simple (unweighted) transfer PageRank baseline.
- Rank Divergence
 - Compute Spearman’s ρ and Kendall’s τ between EndorseRank and AWP.
 - Visualize via scatter plots and top-k Jaccard overlap.
- Computational Efficiency
 - Hardware: Fixed 22-vCPU, 64 GB RAM server.
 - Measures:
 - * Runtime (graph build + PageRank convergence, averaged over 5 runs)
 - * Peak memory usage
 - * Iterations to reach tolerance 10^{-6}

5.2 Data Analysis

1. Hypothesis Testing

- H1 (Divergence)
 - Test whether rank correlations between EndorseRank and AWP differ significantly from 1 using a one-sample t-test or Wilcoxon signed-rank test.
- H2 (Trading Success Alignment)
 - Compare Spearman’s ρ and Kendall’s τ coefficients for EndorseRank vs. AWP against each GMX proxy using paired bootstrap confidence intervals or Fisher z-transform tests for correlated correlations.
- H3 (Computational Efficiency)

- Paired t-tests (or Wilcoxon tests if non-normal) on runtime, memory, and iteration counts.

2. Robustness Checks

- Vary damping factor $d \in \{0.75, 0.85, 0.95\}$.
- Repeat analyses on subgraphs of top ERC-20 tokens by volume on Arbitrum One.
- Assess sensitivity to sample size (50k vs. 200k wallets).
- Compare raw vs. size-weighted realized gain proxy definitions.

3. Reproducibility

- All code, queries, and analysis scripts versioned on GitHub.
- Data snapshots and Docker images preserved for full replication of results.

This methodological framework ensures a rigorous, reproducible evaluation of EndorseRank’s advantages over existing PageRank-based approaches in reputation structure, GMX trading-success alignment, and computational performance.

6 Limitations

While this study is designed to rigorously evaluate EndorseRank against Adaptive Weighted PageRank, we acknowledge several inherent limitations and potential threats to validity:

1. Data Coverage and Quality

- **Event Completeness:** We rely on public event logs (Approve, Permit, Transfer, GMX **PositionDecrease**) from Arbitrum One archives or BigQuery. Some missing or misindexed events (e.g., due to chain reorganizations or node sync issues) may bias graph construction or GMX outcome aggregation.
- **Proxy Limitations:** Trading-success proxies (close-success count, realized gain proxy) are domain-intuition measures, not external ground-truth credit labels. High correlation does not by itself establish predictive validity for unrelated under-collateralized lending outcomes studied in prior lending-focused scoring work (e.g., Ghosh et al., 2024).

2. Temporal Scope

- **Fixed Window:** Limiting analysis to a single six-month window may not capture longer-term reputation dynamics or market regime shifts. Results may differ for other periods, especially during high volatility or GMX protocol upgrades.

3. Graph Modeling Assumptions

- **Approval Semantics:** We assume that any nonzero approve/permit represents a substantive trust endorsement. In practice, users may issue “spend-all” approvals by default or interact via custodial services, which dilute the trust signal.
- **Edge Weight Aggregation:** Summing all transfers or using only the latest allowance ignores the provenance of funds and may obscure multi-token or multi-chain interactions.
- **GMX Trading Semantics:** Profitable **PositionDecrease** closes reflect realized PnL on perpetual swap positions, not loan repayment. The credit analogy is interpretive; wash trading,

copy trading, and automated strategies may inflate proxies without reflecting genuine trustworthiness.

4. Algorithmic Simplifications

- **Fixed Damping Factor:** We apply a single damping factor ($d = 0.85$) uniformly; in reality, the optimal factor may vary by token, wallet type, or network segment.
- **Unmodeled Attributes:** EndorseRank ignores off-chain factors (e.g., KYC status, IP geolocation) and non-ERC-20 relationships (e.g., NFT approvals, cross-chain bridges), which may also influence trust.
- **Minimum Close Filter:** Requiring $\text{total_closes} \geq 3$ excludes novice or inactive traders and may introduce selection bias in the matched sample.

5. Generalizability

- **Protocol Specificity:** We focus on GMX V2 perpetual swap outcomes on Arbitrum One. EndorseRank's alignment may differ for other perpetual exchanges (e.g., dYdX, Hyperliquid) or non-trading use cases.
- **Asset Bias:** High-volume tokens (e.g., USDC, WETH) dominate graph structure; smaller or emerging tokens may yield sparse endorsement networks less amenable to PageRank.

6. Adversarial Manipulation and Sybil Attacks

- **Gaming the Graph:** Malicious actors could issue dummy approvals or orchestrate circular allowance patterns to inflate their EndorseRank score. Similarly, coordinated small profitable closes could inflate close-success count. Detecting and mitigating such Sybil strategies require additional safeguards beyond the scope of this study.
- **No Personhood Layer:** EndorseRank is computed entirely from on-chain state  therefore incorporates none of the off-chain evidence—government-identity checks, biometrics, or web-of-trust vouching—that attestation-based tooling such as Human Passport uses to establish unique personhood (Human Passport, n.d.). It consequently offers no independent

guarantee that two highly ranked addresses are controlled by different actors. This is a deliberate design choice that preserves permissionless participation and avoids personal-data processing, but it means EndorseRank should be read as a relational reliability ordering computed over an already admitted population, not as a Sybil defence in itself.

- **Untested Composition:** Because a personhood gate and a relational ranking answer different questions, combining the two—filtering addresses with an attestation threshold before computing EndorseRank—is a plausible deployment pattern, and the residual weaknesses of attestation scoring such as credential farming would carry over to it (Bordeianu & Popescu, 2026; Siddarth et al., 2020). Evaluating that composition requires labelled Sybil data that this study does not collect, and it is left to future work.

7. Computational Constraints

- **Scalability Limitations:** Although EndorseRank is more efficient than AWP on our sample, real-time computation on the full Arbitrum transfer graph (millions of wallets) may still demand distributed computing or approximation techniques.

7 Ethics Considerations

Although this research relies exclusively on publicly available blockchain data, we will observe the following ethical principles to ensure responsible conduct. Because the study uses non-personal, public ledger data and does not intervene with human participants, it will align with UNISA's research ethics principles on privacy, confidentiality, responsible data management, and prior ethics reviews where required.

Ethics clearance process: before starting the empirical data collection and analysis phase, the researcher will submit the proposal, data-management plan, and public-data justification to the relevant UNISA School ethics process. If the study is classified as exempt or low risk because it uses public pseudonymous blockchain data and no human-participant intervention, the exemption or clearance confirmation will be retained as part of the research record. No attempt will be made to deanonymize wallet owners, combine wallet addresses with off-chain personal data, or publish address-level findings in a way that could reasonably identify individuals.

1. Privacy and Anonymity

- **Pseudonymity Respect:** Arbitrum One addresses are pseudonymous, not inherently linked to real-world identities. We will not attempt to deanonymize addresses or correlate them with personally identifying information.
- **No Sensitive Data Collection:** We will restrict data collection to on-chain events (approve, permit, transfer, GMX **PositionDecrease**) and avoid scraping or integrating off-chain data such as IP addresses or KYC records.

2. Data Usage and Compliance

- **Open Data Sources:** All queries will target public datasets (e.g., Google BigQuery goog_blockchain_arbitrum_one_us) in accordance with their terms of service and rate limits.
- **No Malicious Interaction:** We will not broadcast transactions or otherwise interact with smart contracts in ways that could disrupt network operations or degrade performance.

3. Fairness and Bias Mitigation

- **Algorithmic Fairness:** We will audit EndorseRank and AWP for unintended biases—e.g., systematically undervaluing new entrants or low-volume participants—and discuss these limitations in our analysis.
- **Transparency:** All code, parameter choices, and evaluation metrics will be documented and published, enabling peer review and independent replication.

4. Responsible Disclosure

- **Vulnerability Reporting:** If our analysis uncovers potential vulnerabilities in smart contracts (e.g., unexpected approval patterns that could be exploited), we will confidentially notify the responsible protocol teams before public disclosure.

5. Social and Financial Impact

- **No Financial Advice:** Although EndorseRank may inform risk assessment, we will explicitly disclaim that it is not financial advice and highlight the need for human oversight in lending and trading decisions.
- **Avoiding Misuse:** We will discuss potential adversarial manipulation (e.g., Sybil attacks) and recommend safeguards so that our methodology is not misused to target or disadvantage specific users.

By adhering to these practices, we ensure that our research advances on-chain reputation modeling ethically, respects user privacy, and contributes responsibly to the DeFi ecosystem.

8 Significance of Research

This research advances the field of on-chain reputation scoring and decentralized credit assessment in several meaningful ways. At the doctoral level, the intended original contribution is the formulation and empirical validation of allowance-based wallet reputation as a distinct class of on-chain trust signal. The study is positioned to contribute new knowledge by showing whether explicit authorization relationships can produce a reputation metric that is theoretically interpretable, aligned with GMX V2 perpetual trading success, and computationally more scalable than transaction-history-based PageRank baselines:

1. Theoretical Innovation

- **Endorsement-Based Graph Modeling:** By introducing a directed, weighted endorsement graph derived from ERC-20 approve and permit events, EndorseRank extends the original PageRank citation paradigm to capture explicit on-chain trust signals. This conceptual shift—from token transfers to authorization links—broadens the theoretical underpinnings of network centrality in blockchain contexts.
- **Temporal Simplification:** Unlike existing approaches that require elaborate time-decay functions, EndorseRank leverages the override semantics of ERC-20 allowances to dispense with time-based weighting, offering a cleaner, more principled framework for modeling enduring trust relationships.
- **Leveraged-Exposure Validation Frame:** By validating reputation scores against GMX V2 close-success count and realized gain proxy—rather than transfer in-degree/in-value alone—the study connects graph-based reputation to collateralized perpetual trading outcomes on Arbitrum One.
- **A Distinct Contribution Relative to Attestation-Based Tooling:**  The reputation tooling currently deployed in DeFi, exemplified by Human Passport, scores a wallet by aggregating weighted credentials about its holder and comparing the total against a passing threshold,

drawing on both on-chain and off-chain evidence in order to resist Sybil attacks (Human Passport, n.d.; Bordeianu & Popescu, 2026). That approach answers whether an address is a distinct person. EndorseRank answers a question that remains open once personhood has been established—how wallets should be ordered relative to one another on the strength of the spending authority they delegate to each other—using no off-chain data and producing a continuous ranking rather than an admission decision. The originality claimed here is thus the treatment of delegation as a measurable, rankable trust relation, a signal that neither the graph-based nor the attestation-based prior art uses (Section 3.4).

2. Methodological Contribution

- **Efficient Computation:** EndorseRank demonstrates that a sparser approval graph can yield reputation scores with comparable or superior trading-success alignment to transfer-based PageRank, while reducing computation time, memory usage, and convergence iterations. This efficiency gain makes large-scale, near-real-time reputation scoring more feasible for resource-constrained DeFi platforms.
- **Reproducible Framework:** We provide a fully specified, open-source pipeline—from Arbitrum One data extraction to PageRank implementation and rank-correlation evaluation—enabling other researchers and practitioners to validate, extend, and apply the EndorseRank methodology across different protocols and ecosystems.

3. Practical Impact

- **Improved Risk Management:** By more accurately identifying wallets whose on-chain endorsement structure aligns with successful GMX perpetual trading, EndorseRank can support differentiated risk parameters for collateralized and under-collateralized products. This has the potential to unlock significant liquidity in DeFi and broaden access to financial services.
- **Protocol Integration:** The lightweight nature of EndorseRank makes it a strong candidate for integration into on-chain or off-chain risk oracles, governance dashboards, and compliance

tools, providing actionable intelligence to DeFi protocols, perpetual exchanges, custodial services, and regulators.

4. Broader Implications

- **Cross-Protocol Applicability:** While this study focuses on Arbitrum One and GMX V2, the EndorseRank approach generalizes to any ERC-20-compatible network or custom token standard, making it relevant for emerging L2 solutions and alternative smart-contract platforms.
- **Foundation for Future Research:** By highlighting the value of authorization networks and GMX-derived trading-success proxies, this work encourages further exploration of other on-chain signals—such as governance votes, staking approvals, and cross-chain bridges—and their integration into composite reputation and identity frameworks.

9 Schedule

The following schedule assumes that the research proposal will be completed and approved by 31 August 2026. Post-approval research activities therefore begin in September 2026. Earlier setup, data-acquisition, and modelling milestones are merged into fewer periods so that the thesis submission target for examination remains January 2028. The timeline includes thesis-writing milestones as well as the preparation and submission of journal articles based on the research results.

Table 5: Research Project Schedule

Period	Activities	Deliverables
By 31 August 2026	Complete final proposal revisions Obtain supervisor and departmental approval of the proposal Confirm the approved research scope and post-approval timeline	Approved research proposal Confirmed post-approval work plan
Sep–Oct 2026	Prepare the post-approval research protocol and data-management / ethics-alignment procedures Set up the analysis environment Develop and test the data-acquisition pipeline Extract ERC-20 Approve/Permit and Transfer events on Arbitrum One Extract and decode GMX V2 PositionDecrease events	Approved research protocol Configured research environment ETL scripts Raw and decoded event datasets

Nov–Dec 2026	Clean and validate the extracted datasets Construct wallet-transaction and endorsement graphs Compute trading-success proxies Implement EndorseRank together with AWP and PageRank baselines Run initial parameter-sensitivity checks	Validated analysis and graph datasets Trading-success proxy file Model/baseline implementations Preliminary scoring outputs
Jan–Mar 2027	Evaluate rank divergence and trading-success alignment Compute Spearman’s rho and Kendall’s tau vs. GMX proxies Assess computational efficiency and robustness Interpret results and prepare figures, tables, and chapter-level findings	Evaluation results Alignment / robustness tables Results chapter materials
Apr–May 2027	Draft, revise, and submit the first journal article to an accredited journal Incorporate article findings into the thesis results chapter	Submitted first journal article Updated thesis results chapter

June–August 2027	Draft and revise the introduction, literature review, and methodology chapters Integrate supervisor comments across completed chapters Draft the second publication from the DeFi reputation and GMX alignment findings	Revised thesis chapters Second publication draft Publication-ready tables and evidence
September 2027	Revise and submit the second article to an accredited journal Update the thesis discussion with publication feedback where applicable	Submitted second journal article Updated discussion chapter
October 2027	Complete the full thesis draft Finalize conclusions, limitations, and future-work sections Conduct internal consistency and reference checks	Complete thesis draft Checked references and cross-references
November– December 2027	Revise the full thesis based on supervisor feedback Edit formatting, language, tables, and figures Prepare submission documentation	Final revised thesis Submission-ready document Completed administrative documents

January 2028	Obtain final supervisor sign-off	Submitted thesis
	Submit the thesis for examination	Supervisor sign-off
	Track journal-review outcomes and	Journal response plan
	respond to editorial queries if required	

10 Dissertation/Thesis Outline

The Dissertation consists of the following six chapters:

- Chapter 1: Introduction and research problem. This chapter will refine the proposal into the thesis introduction by presenting the DeFi leveraged-trading context on Arbitrum One, the on-chain reputation score problem, the research questions, objectives, scope, significance, and limitations.
- Chapter 2: Literature review and theoretical framework. This chapter will expand the proposal literature review into a comprehensive analysis of graph-based reputation scoring, DeFi credit-risk prediction, ERC-20 allowance semantics, PageRank variants, and theoretical foundations for reputation structure, GMX trading-success alignment, and computational efficiency.
- Chapter 3: Research methodology. This chapter will specify the philosophical position, research design, data sources, sampling logic, feature engineering, graph-construction procedures, baseline reproduction, statistical tests, robustness checks, ethics clearance, and reproducibility controls.
- Chapter 4: Implementation and empirical results. This chapter will report on the data pipeline, EndorseRank implementation, baseline implementations, dataset characteristics, graph statistics, GMX trading-success alignment results, rank-divergence findings, and computational benchmark outcomes.
- Chapter 5: Discussion. This chapter will interpret the findings against each research question and objective, explain the theoretical and practical implications for DeFi risk assessment, evaluate threats to validity, and discuss how the results support or limit the claimed contribution.
- Chapter 6: Conclusion and future work. This chapter will synthesize the study, state the original contribution to knowledge, revisit the research questions, summarize limitations, and propose future work on cross-protocol validation, adversarial manipulation, and broader on-

chain trust signals.

References

- Arbitrum Foundation. (n.d.). *Arbitrum documentation*. docs.arbitrum.io
- Bordeianu, A. A., & Popescu, D. E. (2026).  Great landscape in decentralized systems: Sybil attacks, related vulnerabilities, and blockchain security evolution (2015–2025). *Applied Sciences*, 16(14), 6929. doi:10.3390/app16146929
- Buterin, V. (2014). *A next-generation smart contract and decentralized application platform*. Ethereum Foundation.
- Buterin, V., Hitzig, Z., & Weyl, E. G. (2019).  flexible design for funding public goods. *Management Science*, 65(11), 5171–5187. doi:10.1287/mnsc.2019.3337
- Chakraborty, S., Aich, S., Sim, J. S., & Kim, H. C. (2019). A blockchain based credit analysis framework for efficient financial systems. In *2019 21st International Conference on Advanced Communication Technology (ICACT)* (pp. 56–60). IEEE.
- Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. *Psychological Bulletin*, 52(4), 281–302.
- Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. *International Journal of Software Innovation*, 11(1), 1–13. doi:10.4018/IJSI.315737
- Ethereum Improvement Proposals. (2015). ERC-20 token standard (EIP-20). EIP-20
- Ethereum Improvement Proposals. (2020). ERC-20 permit extension (EIP-2612). EIP-2612
- Ghosh, R., Datta, A., Sinha, S., Aggarwal, V., & Sengupta, R. (2024). On-chain credit risk score in decentralized finance (arXiv preprint arXiv:2412.00710, revised 2025). arXiv:2412.00710
- GMX. (n.d.). *GMX V2 documentation*. docs.gmx.io
- Hassija, V., Bansal, G., Chamola, V., Kumar, N., & Guizani, M. (2020). Secure lending: Blockchain and prospect theory-based decentralized credit scoring model. *IEEE Transactions on Network Science and Engineering*, 7(4), 2566–2575. doi:10.1109/TNSE.2020.2982488
- Human Passport. (n.d.). *Human Passport developer documentation*.  ps.passport.xyz
- Kotb, M. O. (2023). Credit scoring using machine learning algorithms and blockchain technol-

- ogy. In *2023 Intelligent Methods, Systems, and Applications (IMSA)* (pp. 381–386). IEEE. doi:10.1109/IMSA58542.2023.10217411
- Lin, D., Wu, J., Fu, Q., Zheng, Z., & Chen, T. (2024). RiskProp: Account risk rating on Ethereum via de-anonymous score and network propagation. *IEEE Transactions on Dependable and Secure Computing*, 22(3), 2132–2145. doi:10.1109/TDSC.2024.3475825
- Molnár, B., Pisoni, G., Kherbouche, M., & Zghal, Y. (2023). Blockchain-based business process management (BPM) for finance: The case of credit and claim requests. *Smart Cities*, 6(3), 1254–1278. doi:10.3390/smartcities6030061
- Packin, N. G., & Lev-Aretz, Y. (2024). Decentralized credit scoring: Black box 3.0. *American Business Law Journal*, 61(2), 91–111. doi:10.2139/ssrn.4375920
- Page, L., Brin, S., Motwani, R., & Winograd, T. (1998). *The PageRank citation ranking: Bringing order to the web* (Technical Report). Stanford Digital Library Technologies Project.
- Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. *Federal Reserve Bank of St. Louis Review*, 103(2), 153–174. doi:10.20955/r.103.153-74
- Shimpi, M. (2024). *Credit score-based lending system on the Ethereum Platform* (Master's project No. 1373). San Jose State University. doi:10.31979/etd.kq2w-p3ah
- Siddarth, D., Ivliev, S., Siri, S., & Berman, P. (2020).  Who watches the watchmen? A review of subjective approaches for Sybil-resistance in proof of personhood protocols. *Frontiers in Blockchain*, 3, 590171. doi:10.3389/fbloc.2020.590171
- Szabo, N. (1997). Formalizing and securing relationships on public networks. *First Monday*, 2(9).
- Vathsala V., & V. D. Sharma. (2025). A secure AI architecture for dynamic credit scoring in DeFi environments. *International Journal of Research in Multidisciplinary Technology*, 1(4). ijrmt.com/article/view/4

11 Appendices

None.